CS24303 — THEORY (3.0 CR) FULL COURSE MASTER BIT MESRA | NEP SCHEME

Data Mining Concepts and Techniques (CS24303) — Full Course Master Book

Complete End-to-End B.Tech CSE 5th Semester Study Book & PYQ Bank

Module I: Data Attributes & Proximity Metrics — Topics Covered

1. KDD Process & Data Mining Functionalities
2. Data Object Types & Attribute Categories
3. Five-Number Summary, Boxplots & Dispersion
4. Proximity Metrics: Euclidean, Manhattan, Minkowski
5. Cosine Similarity & Jaccard Coefficient

1. Proximity and Distance Metrics Formulations

Metric Name	Mathematical Formulation	Application Domain
Euclidean Distance (L_2)	$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$	Standard spatial coordinate distance for continuous numeric features.
Manhattan Distance (L_1)	$d(x, y) = \sum_{i=1}^{n} \lvert x_i - y_i \rvert$	Grid-like city block routing; robust against extreme outliers.
Cosine Similarity	$\text{sim}(x, y) = \frac{x \cdot y}{\lVert x \rVert \lVert y \rVert} = \frac{\sum x_i y_i}{\sqrt{\sum x_i^2}\sqrt{\sum y_i^2}}$	Document text mining, TF-IDF vector comparisons (invariant to document length).
Jaccard Coefficient	$J(A, B) = \frac{\lvert A \cap B \rvert}{\lvert A \cup B \rvert} = \frac{q}{q + r + s}$	Asymmetric binary attributes (e.g., market basket item presence).

Module II: Data Preprocessing & Normalization — Topics Covered

1. Data Cleaning & Smoothing by Binning
2. Correlation Analysis: Chi-Square (χ^2) & Pearson r
3. Min-Max Normalization & Z-Score Scaling
4. Dimensionality Reduction via PCA
5. Discretization & Concept Hierarchies

1. Data Transformation: Normalization Formulations

Normalization Method	Formula	Key Use Case
Min-Max Normalization	$v' = \frac{v - \min_A}{\max_A - \min_A}(new\_\max_A - new\_\min_A) + new\_\min_A$	Maps values linearly into predetermined interval $[0, 1]$.
Z-Score (Zero-Mean)	$v' = \frac{v - \mu_A}{\sigma_A}$	Useful when min and max are unknown or outliers dominate.

Module III: Data Warehousing & OLAP Cubes — Topics Covered

1. OLTP vs. OLAP Multidimensional Modeling
2. Schemas: Star, Snowflake & Fact Constellation
3. Lattice of Cuboids & Data Cube Computation
4. OLAP Operations: Roll-up, Drill-down, Slice, Dice, Pivot
5. Attribute-Oriented Induction (AOI)

1. Core OLAP Operations

- **Roll-Up (Drill-Up):** Summarizes data by climbing up concept hierarchies (e.g., Day → Month → Quarter → Year).
- **Drill-Down (Roll-Down):** Navigates from higher-level summary to detailed granular data.
- **Slice:** Performs a selection on one dimension (e.g., `Time = "Q1"`).
- **Dice:** Defines a sub-cube by selecting on two or more dimensions (e.g., `Time = "Q1"` and `Location = "India"`).
- **Pivot (Rotate):** Rotates multidimensional axes to provide an alternative perspective.

Module IV: Apriori & FP-Growth Pattern Mining — Topics Covered

1. Market Basket Analysis: Support & Confidence
2. The Apriori Downward Closure Property
3. Apriori Candidate Generation: Join & Prune
4. FP-Tree Construction & Recursive Mining
5. Pattern Correlation: Lift & Kulczynski

1. Support, Confidence & Lift Formulations

$$\text{Support}(A \implies B) = P(A \cup B) = \frac{\text{count}(A \cup B)}{|D|}$$

$$\text{Confidence}(A \implies B) = P(B \mid A) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

$$\text{Lift}(A, B) = \frac{P(A \cup B)}{P(A)P(B)}$$

🏛 **BIT Mesra Exam Question (10 Marks)**

Problem: Explain the Apriori property and demonstrate how the prune step eliminates candidate itemsets without scanning the database.

Solution: *Apriori Property:* All non-empty subsets of a frequent itemset must also be frequent. If any $(k-1)$-subset of candidate C_k is not present in frequent list L_{k-1}, C_k cannot be frequent and is immediately pruned without costly database I/O.

Module V: Advanced Pattern Mining & Applications — Topics Covered

1. Multilevel & Multidimensional Association Rules
2. Quantitative Association Rule Mining
3. Constraint-Based Mining (Antimonotonic, Monotonic, Succinct)
4. Mining Colossal & Compressed Patterns
5. Real-World Applications (Retail Basket, Fraud Detection)

1. Constraint Properties in Frequent Itemset Mining

Constraint Type	Formal Property Definition	Pushing Strategy in Mining
Antimonotonic	If itemset S violates constraint C, all supersets $S' \supset S$ also violate C. (e.g., $\text{sum}(S.\text{price}) \leq 100$).	Prunes itemset search branch immediately (like Apriori support).
Monotonic	If itemset S satisfies constraint C, all supersets $S' \supset S$ also satisfy C. (e.g., $\text{sum}(S.\text{price}) \geq 500$).	Once satisfied, all descendant supersets need not be tested against C.